4 **(a)** A restaurant offers a fixed priced menu. A meal for an adult costs $9.99 and a meal for a child costs $6.99.

If there are 6 or more people at a table, a 15% discount is applied to the total cost of the order.

Write an algorithm in pseudocode to:

- input the number of adults and the number of children at a table
- calculate the total cost of the order with the discount, if this applies
- output the final cost of the order.

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..... [5]

(b) The final cost of the order after the discount may have more than 2 decimal places.

Write a pseudocode statement to set the final cost of the order to 2 decimal places.
You do **not** need to rewrite the whole algorithm.

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..... [2]

(c) The restaurant offers a discount on a meal if the person ordering is either a student or is 65 years or older. The discount (**X** = 1) can only be given if the amount of the order is $20.00 or over.

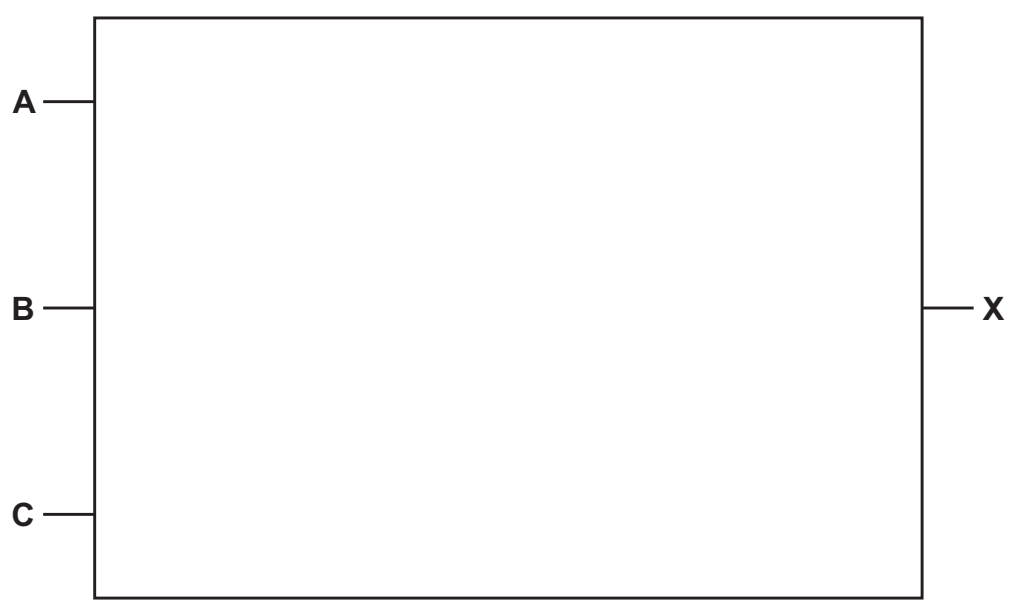
rule **A**	a student	0
	not a student	1
rule **B**	under 65 years old	0
	65 years or older	1
rule **C**	under $20.00	0
	$20.00 or over	1

(i) Write the logic expression for this problem.

..... [2]

(ii) Draw a logic circuit for this logic expression.

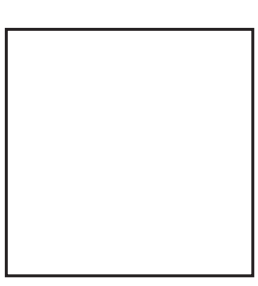
Each logic gate must have a maximum of **two** inputs.
Do **not** simplify this logic expression.



[3]

(d) Draw the NAND logic gate symbol **and** complete the truth table for it.

NAND logic gate symbol:



Truth table:

A	B	X
0	0	
0	1	
1	0	
1	1	

[2]